Database Diagram (dùng từ Question 2 trở đi)

Các bảng có trong database:

staffs: staff_id, first_name, last_name, phone, active, store_id, manager_id

stores: store_id, store_name, phone, email, street, city, state, zip_code

orders: order_id, customer_id, order_status, order_date, required_date, shipped_date, store_id, staff_id

stocks: store_id, product_id, quantity

customers: customer_id, first_name, last_name, phone, email, street, city, state, zip_code
order_items: order_id, item_id, product_id, quantity, list_price, discount

products: product_id, product_name, model_year, list_price, brand_name, category_name

Question 1

[1.5 marks] Create a database and then write SQL statements to create, inside it, all the tables derived from the ERD shown in the following picture, with appropriate attributes, primary keys and foreign keys.

NOTE that when writing the SQL commands as requested, you MUST keep the names of tables, relationships, attributes and data types of attributes exactly the same as those given in the ERD.

Attributes with an underline belong to the Primary Key of each entity.

Attributes that reference the primary key of another table must have the same name as the primary key attributes of that table.

When submitting the response for this question, include only the SQL statements that create the tables with their corresponding keys and foreign keys. Do not use "create database", "Alter database", "use database_name" or any statement that mentions a database name in your submission.

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Entities và attributes:

Product
  - ProductID: int  [PK, underlined]
  - Productname: nvarchar(150)
  - Price: decimal(12,2)

Category
  - CategoryID: int  [PK, underlined]
  - Name: nvarchar(100)
  - Description: nvarchar(200)

Supplier
  - SupplierID: varchar(30)  [PK, underlined]
  - SupplierName: nvarchar(100)
  - Level: int

Relationships:
  - Product "has" Category  (Product N – 1 Category)
  - Product "hasSupplier" Supplier  (Product N – M Supplier)
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Question 2

[0.5 mark] Write a query that lists the customers living in New York. The result includes the columns customer_id, first_name, last_name, email, sorted in ascending order by last_name and first_name as follows.

customer_id	first_name	last_name	email
1	Liam	Johnson	liam.johnson@cust.example
2	Emma	Smith	emma.smith@cust.example

Question 3

[1 mark] Write a query that lists the products whose product_name contains the word 'City'. The result includes the columns product_id, product_name, and list_price, sorted in ascending order by product_name as follows.

product_id	product_name	list_price
6	City Lite	259.00
7	City Pro	399.00
11	E-City 250	1299.00

Question 4

[1 mark] Write a query that finds orders between 2025-08-05 and 2025-08-15 (YYYY-MM-DD). Display the results with the columns order_id, customer_id, order_date, store_id, staff_id, sorted in ascending order by the order_date column as follows.

order_id	customer_id	order_date	store_id	staff_id
5	5	2025-08-05	5	5
6	6	2025-08-06	6	6
7	7	2025-08-07	7	7
8	8	2025-08-08	8	8
9	9	2025-08-09	9	9
10	10	2025-08-10	10	10
11	11	2025-08-11	11	11
12	12	2025-08-12	12	12
13	13	2025-08-13	13	13
14	14	2025-08-14	14	14
15	15	2025-08-15	15	15

Question 5

[1 mark] Write a query that finds the products having the lowest list_price. Display the results with the columns product_id, product_name, and list_price as follows.

product_id	product_name	list_price
10	Kids Fun 14	149.00
13	BMX Spark	149.00

Question 6

[1 mark] Write a query that calculates the total quantity in each store. The result includes the columns store_id, store_name, and total_qty_in_stock, sorted in descending order by total_qty_in_stock and in ascending order by store_id as follows.

store_id	store_name	total_qty_in_stock
1	Downtown Cycles	17
2	Riverside Wheels	16
4	Desert Trail	13
7	Seaside Spokes	13
3	Bay City Bikes	11
6	Music City Pedals	11
5	Capitol Ride	9

Question 7

[1 mark] Write a query that finds the highest list_price of the products within each category_name. The result includes the columns category_name, product_id, product_name, and list_price, sorted in ascending order by category_name and product_id as follows.

category_name	product_id	product_name	list_price
BMX	13	BMX Spark	149.00
City	7	City Pro	399.00
E-Bike	12	E-Mountain 500	2199.00
Fixie	15	Fixie Basic	219.00
Folding	9	Folding Go	379.00
Gravel	8	Gravel GX	999.00
Hybrid	17	Hybrid H3	649.00
Kids	10	Kids Fun 14	149.00
Mountain	5	Mountain X2	799.00
Road	3	Roadster 300	699.00
Touring	14	Touring T1	899.00